

SiO₂ Growth & Diffusion

Interactive 3D Furnace Simulator

🔥 FURNACE HOT ZONE 🔥

4 Process Modes

Deal-Grove Physics

Fick's 2nd Law

Real-time 3D

Gas Flow Simulation

Download the HTML from GitHub to begin ([Link](#))

Interface Overview

Three panels + a live 3D viewport fill the screen

LEFT PANEL

Physics & Metrics

- Deal-Grove coefficients A, B
- Gas flow physics (Re , λ , Kn)
- Thermodynamics (kT , T in K)
- Live growth / diffusion charts
- Gas species legend

3D VIEWPORT

Interactive WebGL Canvas

- Quartz furnace tube + wafer boat
- Animated gas particles
- Oxide layers grow in real time
- Orbit · Zoom · Pan with mouse
- Click wafer → tooltip info

RIGHT PANEL

Controls

- Process type selector
- Temperature & gas-flow sliders
- Time acceleration slider
- Target thickness / depth
- Simulation control buttons

🕒 **BOTTOM BAR:** Process Time | Oxide progress bar | Furnace temperature readout

Step 1 — Choose Your Process Type

Right panel → Process Type — four modes, each with distinct physics

DRY OXIDE (Dry O₂)

Reaction: $\text{Si} + \text{O}_2 \rightarrow \text{SiO}_2$
Pure O₂ thermal oxidation
Densest, highest-quality oxide
Lowest interface-state density (Dit
 $\sim 10^{10} \text{ cm}^{-2} \text{ eV}^{-1}$)
Used for gate dielectrics, pad oxide
Slower growth rate vs wet
Gas: 50% O₂ + 50% N₂ carrier

WET OXIDE (H₂O)

Reaction: $\text{Si} + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 2\text{H}_2\uparrow$
Steam (pyrogenic H₂O) oxidation
5–10× faster growth than dry
Lower density, higher Dit
Used for thick field oxide, LOCOS
H₂O diffuses faster through SiO₂
Gas: 55% H₂O + 30% O₂ + 15% H₂

P-DIFF (Boron)

Reaction: $\text{BCl}_3 + \text{O}_2 \rightarrow \text{B}_2\text{O}_3 \rightarrow \text{B}\downarrow$
Boron = acceptor → p-type regions
Lower activation energy (E_a = 3.46 eV)
Fast diffuser in silicon
Creates p⁺ source/drain, base
BSG acts as dopant source layer
Gas: 5% BCl₃ + 25% O₂ + 70% N₂

N-DIFF (Phosphorus)

Reaction: $4\text{POCl}_3 + 3\text{O}_2 \rightarrow 2\text{P}_2\text{O}_5$
Phosphorus = donor → n-type regions
Activation energy E_a = 3.66 eV
POCl₃ liquid source, PSG forms
Also getters metallic contaminants
Creates n⁺ emitter, buried layers
Gas: 5% POCl₃ + 25% O₂ + 70% N₂

Step 2 — Furnace Parameter Controls

Right panel → Furnace Parameters — five sliders with live preview

Temperature

Range: 600 – 1200 °C Default: 1000 °C

- Controls reaction rate via Arrhenius law
- Higher T → exponentially faster oxide/diffusion
- Affects Deal-Grove B and B/A coefficients
- Also shown live in bottom-right readout (orange)

Gas Flow Rate

Range: 0.5 – 10 SLM Default: 2.0 SLM

- Sets standard liters per minute into tube
- Affects Reynolds number (laminar vs turbulent)
- Changes particle speed in 3D animation
- Higher flow → steeper parabolic velocity profile

Time Acceleration

Range: 1× – 500× Default: 20×

- Speeds up simulated time relative to real time
- 1× = real-time (very slow for thick oxides)
- 20× default gives visible growth in seconds
- 500× useful for reaching thick oxide targets fast

Particle Density

Range: 10% – 100% Default: 100%

- Controls how many gas particles are animated
- Lower % reduces visual clutter for screenshots
- No effect on physics — purely visual
- 10% still shows flow pattern clearly

Max Process Time

Range: 0.25 – 24 hr Default: 2.0 hr

- Sets the process window / x-axis of growth charts
- Type exact value in number input field
- Enable Stop at max to auto-pause simulation
- Progress bar in bottom tracks time vs max

Step 3 — Setting a Target & Auto-Stop

Right panel → Target Thickness/Depth — goal-driven process engineering

OXIDE TARGET (Dry / Wet modes)

Slider + number input: 1 – 2000 nm

Sets the target SiO_2 thickness you want to grow.

Progress bar fills from blue → yellow → green as the simulation approaches your target.

✓ "Target reached — simulation paused" message appears in green when thickness is hit.

Toggle "■ Auto-stop" to automatically pause the simulation the moment the target is reached.

Tip: set a thin target (e.g. 10 nm) with Auto-stop ON to see exactly when gate-oxide quality is met.

JUNCTION DEPTH TARGET (P-Diff / N-Diff)

Slider + number input: 0.1 – 20 μm

Sets the target p-n junction depth x_j in the silicon substrate you want to reach.

The simulator solves Fick's 2nd Law to compute x_j = depth where $C(x,t)$ = background doping.

Same progress bar and Auto-stop logic applies.

Toggle "■ Auto-stop" to pause when x_j is met.

Tip: N-Diff (Phosphorus) requires higher temps to match Boron junction depths — compare both!

Step 4 — Simulation Controls & Camera

Right panel → Simulation Control + Camera Views

II / ▶ Pause / Run

Toggles the simulation on or off. Button turns blue (highlighted) while running. Press Pause mid-run to inspect current values, then Resume to continue.

↺ Reset

Returns simulation time to 0:00:00, hides all grown oxide layers, and restores wafer thickness to original. Target-reached state is also cleared.

↗ Velocity Field

Overlays 3 animated blue cone arrows at the tube inlet showing parabolic laminar-flow lanes. Useful for visualizing how gas velocity varies across the tube radius.

⦿ Particles ON/OFF

Toggles the animated gas particle cloud. Turn OFF for a cleaner view of oxide layers or when doing a screenshot. Physics and charts still update.

🔄 Auto-Rotate ON/OFF

Continuously rotates the camera around the furnace. Great for presentations or passive demos. Orbit control still works while auto-rotate is on.

🔗 Cross-Section ON/OFF

Repositions the camera to face straight down the tube axis (front view), revealing the circular wafer cross-sections inside the furnace.

📷 CAMERA VIEWS

Isometric

Default perspective view — 22°, 13°, 32° — shows tube, wafers, and inlet clearly.

Side

Pure side view along Z-axis — see the full tube length and wafer boat spacing.

Top

Bird's eye from above — look straight down into the open furnace tube.

End (Front)

Face-on view from the outlet end — see wafer discs in cross-section.

Left Panel Deep-Dive — Deal-Grove Growth Model

Shown in left panel when Dry or Wet mode is active

THE DEAL-GROVE EQUATION

$$x^2 + A \cdot x = B \cdot (t + \tau)$$
$$x = [-A + \sqrt{A^2 + 4Bt}] / 2$$

$$B = B_0 \cdot \exp(-E_b / kT) \quad [\text{parabolic rate}]$$
$$B/A = (B/A)_0 \cdot \exp(-E_{ba} / kT) \quad [\text{linear rate}]$$

PHYSICAL INTERPRETATION

Parabolic regime (thick oxide, long time): B dominates.
Oxide grows as $\sqrt{t}$ — O_2/H_2O must diffuse through existing SiO_2 .

Linear regime (thin oxide, early time): B/A dominates.
Growth is reaction-rate limited at the Si/ SiO_2 interface.

τ accounts for any initial oxide already present (native oxide).

LIVE PARAMETERS (update as you drag sliders)

A — linear coefficient	μm	Ratio B/linear_rate
B — parabolic coefficient	$\mu m^2/hr$	Arrhenius: $B_0 \cdot \exp(-E_b/kT)$
B/A — linear rate	$\mu m/hr$	Dominates thin-oxide regime
Oxide thickness x	nm	Current SiO_2 grown
Growth rate dx/dt	nm/min	Instantaneous rate
Thin-film colour	Colour name	Interference colour (like a wafer!)

LEFT PANEL MINI-CHART — Oxide Thickness vs Time

The canvas labelled Oxide thickness (nm) vs process time updates in real time as the simulation runs.

X-axis: 0 → Max Process Time (set by slider)
Y-axis: 0 → current maximum oxide thickness

Dry (O_2) curve: lower, slower — parabolic shape
Wet (H_2O) curve: higher, faster — same shape, larger B

Dry parameters (from Jaeger / Wolf & Tauber literature):
 $B_0 = 772 \mu m^2/hr$, $E_b = 1.23 \text{ eV}$
 $(B/A)_0 = 6.23 \times 10^6 \mu m/hr$, $E_{ba} = 2.00 \text{ eV}$

Wet parameters:
 $B_0 = 386 \mu m^2/hr$, $E_b = 0.78 \text{ eV}$
 $(B/A)_0 = 1.63 \times 10^8 \mu m/hr$, $E_{ba} = 2.05 \text{ eV}$

Left Panel Deep-Dive — Diffusion Physics

Shown in left panel when P-Diff (Boron) or N-Diff (Phosphorus) mode is active

FICK'S 2ND LAW OF DIFFUSION

$$\frac{\partial C}{\partial t} = D \cdot \frac{\partial^2 C}{\partial x^2}$$
$$C(x,t) = C_s \cdot \operatorname{erfc}\left(\frac{x}{2\sqrt{Dt}}\right)$$
$$D(T) = D_0 \cdot \exp\left(-\frac{E_a}{kT}\right) \quad [\text{Arrhenius}]$$
$$x_j = 2\sqrt{Dt} \cdot \operatorname{erfc}^{-1}(C_b/C_s)$$

UNDERSTANDING THE VARIABLES

$C(x,t)$ — dopant concentration at depth x , time t
 C_s — surface concentration (set by source gas flow)
 D — diffusivity [cm^2/s], increases exponentially with T
 erfc — complementary error function (gives the profile shape)
 x_j — junction depth: where $C(x,t) = \text{background doping } C_b$
 $\sqrt{Dt}$ — diffusion length, key figure of merit

DOPANT PARAMETERS

Species	Boron (B)	Phosphorus (P)
Mode	P-Diff	N-Diff
D_0 (pre-exp)	$0.76 \text{ cm}^2/\text{s}$	$3.85 \text{ cm}^2/\text{s}$
E_a (activation)	3.46 eV	3.66 eV
C_s (surface conc)	$5 \times 10^{19} \text{ cm}^{-3}$	$1 \times 10^{20} \text{ cm}^{-3}$
Source gas	BCl_3	POCl_3
Creates	p-type (acceptor)	n-type (donor)

LEFT PANEL MINI-CHARTS — Diffusion Mode

Chart 1 — Junction depth x_j (μm) vs process time
Shows how fast the dopant front moves into silicon.
 x_j grows as $\sqrt{t}$ (characteristic Fickian behaviour).

Chart 2 — $\log_{10} C(x,t)$: concentration vs depth
Semi-log plot of the dopant profile at current time.
Vertical cursor marks the junction depth x_j .
Shape is erfc — steep near surface, long tail.

Live values shown: D , D_0 , E_a , $\sqrt{Dt}$, x_j
All update in real time as T or time changes.

Gas Flow Physics & 3D Scene Guide

Left panel → Gas Flow Physics + Thermodynamics + Gas Species | 3D viewport interactions

GAS FLOW METRICS (left panel)

Flow rate	Slider value in SLM (std L/min)
Avg velocity	Mean gas speed across tube cross-section
v_max	Centre-line velocity = 2 × avg (parabolic profile)
Reynolds number	$Re = \rho v D / \mu$ — indicates laminar ($Re < 2300$) or turbulent
Flow regime	Always laminar at furnace conditions — uniform delivery
Gas viscosity μ	Sutherland's law mixture viscosity at current T
Mean free path λ	$\lambda = kT / (\sqrt{2} \cdot \pi \cdot d^2 \cdot P)$ — molecular spacing
Knudsen number Kn	$Kn = \lambda / D$ — ratio of λ to tube diameter

THERMODYNAMICS (left panel)

Temperature (°C) — set by slider; matches furnace readout
Temperature (K) — $T(K) = T(^{\circ}C) + 273.15$
Thermal energy kT — in eV; governs Arrhenius rates
Boltzmann $k = 8.617 \times 10^{-5}$ eV/K (constant)

Gas Species panel — colour-coded dot legend for each gas in the current process mixture.
Hover a particle in 3D to see its species tooltip.

3D VIEWPORT — WHAT YOU SEE

Quartz furnace tube	Semi-transparent cylinder, 200 mm ID — see inside
Heating coil	22-turn resistive spiral glowing red/orange outside tube
Wafer boat	9 × 150 mm Si(100) wafers, 2.6-unit spacing
Gas inlet/outlet	Cone + pipe at each end; animated blue flow arrows
Oxide layers	Blue-tinted discs grow on wafer faces over time
Diffusion layers	Orange-tinted discs fade in during P/N-Diff modes
Gas particles	Colour-coded spheres follow parabolic laminar profile
Hot-zone gradient	Inner tube glows red-orange at centre, cool blue at ends

 Mouse Controls: Left-drag = Orbit · Scroll = Zoom · Right-drag = Pan · Click wafer = Tooltip

Quick-Start Workflow

Follow these steps to run your first simulation end-to-end

1

Open the File

Open index.html in any modern browser (Chrome, Edge, Firefox).
The 3D furnace loads immediately — no install needed.
Simulation starts running automatically at 20× speed.

2

Choose a Process

Right panel → Process Type → click one of the four tabs:
Dry O₂ · Wet H₂O · P-Diff (Boron) · N-Diff (Phosphorus)
The badge in the top bar updates. Gas legend on left changes.

3

Set Temperature

Right panel → Furnace Parameters → Temperature slider.
Try 800°C for slow growth, 1100°C for fast. Watch left panel
coefficients (A, B, D) update in real time.

4

Set a Target

Right panel → Target Thickness/Depth → drag slider or type.
Enable ☐ Auto-stop. Watch the progress bar fill toward 100%.
Simulation pauses with ✓ confirmation when target is hit.

5

Explore the 3D View

Left-drag to orbit, scroll to zoom, right-drag to pan.
Use Camera View buttons for preset angles.
Enable  Cross-Section or  Auto-Rotate for presentations.

6

Read the Physics

Left panel shows all live calculated values.
Hover gas particles for species info. Click a wafer for details.
Learning Notes section (right panel, bottom) summarises key concepts.

Tips, Tricks & Learning Experiments

Get the most out of the simulator — experiment-driven learning

Compare Dry vs Wet Growth

- Set $T=1000^{\circ}\text{C}$, Max Time = 2 hr.
- Run Dry O_2 — note oxide at 1 hr mark.
- Reset → Switch to Wet H_2O — same settings.
- Wet is 5–10× thicker at same time → confirms theory.

Observe Arrhenius Scaling

- Set process to Dry O_2 , Auto-stop at 100 nm.
- Run at 900°C , note time-to-target.
- Reset, change to 1100°C , run again.
- Every 100°C roughly doubles/triples growth rate.

Parabolic vs Linear Regime

- Set Max Time to 0.5 hr, $T=1100^{\circ}\text{C}$.
- Watch the mini-chart early slope (linear B/A regime).
- Extend Max Time to 6 hr — curve flattens (parabolic B).
- B/A dominates thin oxide, B dominates thick.

Gate Oxide Precision

- Switch to Dry O_2 . Target = 8 nm, enable Auto-stop.
- Set $T=850^{\circ}\text{C}$ (low — for control).
- This is realistic gate oxide engineering!
- Try 1050°C — same target but much faster.

Boron vs Phosphorus Depth

- P-Diff vs N-Diff at same T and time.
- $E_a(\text{B}) = 3.46 \text{ eV}$ vs $E_a(\text{P}) = 3.66 \text{ eV}$.
- Boron diffuses deeper at same conditions.
- Increase T — gap narrows (both Arrhenius).

Cross-Section + Auto-Rotate

- Enable  Cross-Section → camera snaps to tube end.
- Enable  Auto-Rotate for a 360° fly-around.
- Turn off Particles () for a clean oxide-layer view.
- Perfect for slides, screenshots, or lab demos.

Quick Reference Summary

Everything at a glance — print or bookmark this slide

CONTROLS AT A GLANCE

PROCESS TABS (right panel top)

· Dry O₂ · Wet H₂O · P-Diff (B) · N-Diff (P)

SLIDERS (right panel)

· Temperature: 600–1200°C
· Gas Flow: 0.5–10 SLM
· Time Accel: 1×–500×
· Particle Density: 10–100%
· Max Time: 0.25–24 hr

BUTTONS (right panel)

·   Pause/Run ·  Reset
·  Velocity Field ·  Particles
·  Auto-Rotate ·  Cross-Section

CAMERA PRESETS

· Isometric · Side · Top · End

MOUSE (3D viewport)

· L-drag=Orbit Scroll=Zoom R-drag=Pan
· Click wafer = info tooltip
· Hover particle = species tooltip

BOTTOM BAR

· Process time clock
· Oxide / junction progress bar
· Furnace temp readout (orange)

PHYSICS QUICK REFERENCE

DEAL-GROVE (Dry/Wet)

· $x^2 + Ax = B(t+\tau) \rightarrow x = [-A + \sqrt{A^2 + 4Bt}] / 2$
· $B = B_0 \cdot \exp(-E_b/kT)$ $B/A = (B/A)_0 \cdot \exp(-E_{ba}/kT)$
· Dry: $B_0=772$, $E_b=1.23$ eV
· Wet: $B_0=386$, $E_b=0.78$ eV

DIFFUSION (P/N-Diff)

· $C(x,t) = C_s \cdot \operatorname{erfc}(x / \sqrt{2Dt})$
· $D = D_0 \cdot \exp(-E_a/kT)$
· Boron: $D_0=0.76$, $E_a=3.46$ eV
· Phos: $D_0=3.85$, $E_a=3.66$ eV

GAS FLOW

· $Re = \rho v D / \mu$ (laminar when $Re < 2300$)
· $v(r) = v_{\max} \cdot (1 - r^2/R^2)$ parabolic
· $\lambda = kT / (\sqrt{2} \cdot \pi \cdot d^2 \cdot P)$ mean free path

TUBE / WAFER SPECS

· Tube: 200 mm ID, fused quartz, 1 atm
· Wafer: Si(100) p-type, 150 mm, 650 μm
· 9 wafers in boat · SiO₂/Si ratio: 0.456 nm Si / nm SiO₂